Algebra 3

QUADRATIC EQUATION

Quadratic is an expression or an equation that contains the variable squared, but not raised to any higher power.

The general quadratic equation is expressed as:

$$Ax^2 + Bx + C = 0$$

where, A, B and C are real numbers and with $A \pm 0$.

When B = 0, quadratic equation is known as a **pure quadratic equation**.

A quadratic equation in x is also known as a **second-degree polynomial equation**.

The solution to a quadratic equation is either by factoring or by the use of the quadratic formula.

The following is the quadratic formula:

$$x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

The quantity $\sqrt{B^2 - 4AC}$ in the above equation is known as the **discriminant**.

The discriminant will determine the nature of the roots of the quadratic equation.

The table below shows the value of the discriminant and its corresponding nature of roots.

Descriminant	Nature of roots
0	Only one root (Real and equal)
> 0	Real and unequal
< 0	Imaginary and unequal

SUM & PRODUCT
of roots of Quadratic Equations

The sum and product of the roots of a quadratic equation can be solved even without using factoring or quadratic formula as long as the general equation is given.

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The following are the properties of the roots of a quadratic equation:

Let r_1 and r_2 be the roots of a quadratic equation:

Sum of the roots:
When the two roots are added, the result is:

$$r_1 + r_2 = -\frac{B}{A}$$

Product of the roots:
When the two roots are multiplied, the result is:

$$r_1 \cdot r_2 = \frac{C}{A}$$

Binomial Theorem

Binomial is an expression containing two terms joined by either + or -.

Binomial theorem gives the result of raising a binomial expression to a certain power. The expansion and the series lead to are called the **binomial expansion** and the **binomial series**, respectively.

The binomial theorem is expressed as follows:

$$(x+y)^n = x^n + nx^{n-1}y + \frac{n(n-1)}{2!}x^{n-2}y^2 + \cdots + nxy^{n-1} + y^n$$

The r^{th} term of the binomial expansion of $(x+y)^n$ may be calculated using the following formulas:

$$r^{th} = \frac{n(n-1)(n-2)\cdots(n-r+2)}{(r-1)!}x^{(n-r+1)}y^{r-1}$$

$$r^{th} = {}_nC_{r-1}x^{(n-r+1)}y^{r-1}$$

A term involving a variable with a specific exponent is obtained by using the following formula:

$$y^r = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}x^{n-r}y^r$$

Sum of the coefficients of the expansion of $(x+y)^n$:

$$\text{Sum} = (\text{Coeff. of x} + \text{coeff. of y})^n$$

Sum of exponents of the expansion of $(x+y)^n$:

$$\text{Sum} = n(n+1)$$

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Properties of Binomial Expansion of $(x+y)^n$

1. The number of terms in the resulting expansion is equal to n + 1.
2. The exponent of x decreases by 1 in succeeding terms, while that exponent of y increases by 1 in succeeding terms.
3. The sum of the exponents of each term is equal to n.
4. The first term is x^n and the last term is y^n and each of the terms has a coefficient of 1
5. The coefficient increases and then decreases in a symmetric pattern.

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PASCAL'S TRIANGLE

The coefficients of a binomial expansion can also be conveniently obtained by arranging them in a triangular array or pattern. This is known as Pascal's Triangle named after the famous French mathematician **Blaise Pascal** (1623 – 1662).

Each number in the triangle is equal to the sum of the two numbers immediately above it.

In Italy, this triangular pattern is known as **Tartaglia's triangle** while in many parts of Asia, it is referred to as **Yang Hui's triangle.**

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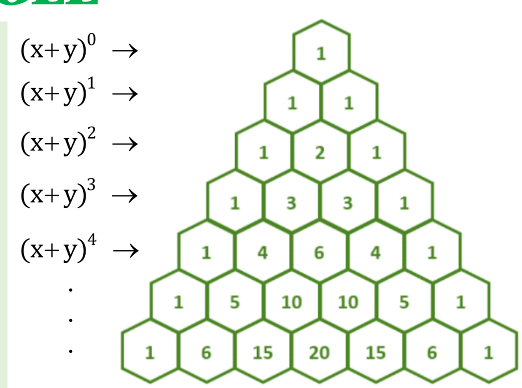

Coefficient of any term in the binomial expansion:

$$C = \frac{(\text{Coeff. of PT})(\text{Exponent of x of PT})}{\text{Exponent of y of PT} + 1}$$